WARNING SIGNS

Table 2. Standard Warning Sign Sizes (page 1 of 2)

Sign	Sign Code	Non-Freeway	Expressway	Freeway
Horizontal Alignment	W1-1, 2, 3, 4, 5	36 x 36	36 x 36	48 x 48
One-Direction Large Arrow	W1-6	48 x 24	48 x 24	60 x 30
Two-Direction Large Arrow	W1-7	48 x 24	48 x 24	60 x 30
Chevron Alignment	W1-8	24 x 30	36 x 48	Ramp 45-55 mph 36 x 48 Ramp 40 mph or less 24x30
Combination Horizontal Alignment/ Intersection	W1-10, 10a, 10b, 10c, 10d, 10e	36 x 36	36 x 36	48 x 48
Hairpin Curve	W1-11	36 x 36	36 x 36	48 x 48
Truck Rollover	W1-13	36 x 36	36 x 36	48 x 48
270 Degree Loop	W1-15	36 x 36	36 x 36	48 x 48
Intersection Warning	W2-1, 2, 3, 4, 5, 6, 7, 8	36 x 36	36 x 36	-
Advanced Traffic Control	W3-1, 2, 3	36 x 36	36 x 36	48 x 48
Reduced Speed Limit Ahead	W3-5	36 x 36	36 x 36	48 x 48
XX MPH Speed Zone Ahead	W3-5a	36 x 36	36 x 36	48 x 48
Reduced Speed Zone Ahead	W3-5b	36 x 36	36 x 36	48 x 48
Merging Signs	W4-1, 2, 3	36 x 36	36 x 36	48 x 48
Cross Traffic Does Not Stop (plaque)	W4-4P, 4aP, 4bP	30 x 15	36 x 18	-
Entering Roadway Merge	W4-5	36 x 36	36 x 36	48 x 48
Entering Roadway Added Lane	W4-6	36 x 36	36 x 36	48 x 48
Road Narrows	W5-1	36 x 36	36 x 36	48 x 48
Narrow Bridge	W5-2	36 x 36	36 x 36	48 x 48
Divided Highway	W6-1	36 x 36	36 x 36	-
Divided Highway Ends	W6-2	36 x 36	36 x 36	-
Two Way Traffic	W6-3	36 x 36	36 x 36	-
Triple Arrow	W6-5	36 x 36	36 x 36	48 x 48
Hill	W7-1, 1a	36 x 36	36 x 36	48 x 48
Bump or Dip	W8-1, 2	36 x 36	36 x 36	48 x 48
Bridge Ices Before Road	W8-13	36 x 36	36 x 36	48 x 48
Left (Right) Lane Ends	W9-1	36 x 36	36 x 36	48 x 48
Lane Ends Merge Left (Right)	W9-2	36 x 36	36 x 36	48 x 48

Table 3. Minimum Advance Sign Placement Distance (ft)

Posted Speed (mph)	Condition A[1]	Condition B (mph)														
		0[2]	5[3]	10[3]	15[3]	20[3]	25[3]	30[3]	35[3]	40[3]	45[3]	50[3]	55[3]	60[3]	65[3]	70[3]
25	325	155	125	100[4]	100[4]	100[4]										
30	460	200	150	115	100[4]	100[4]	100[4]									
35	565	250	185	125	115	100[4]	100[4]	100[4]								
40	670	305	225	150	135	125	120	115	100[4]							
45	775	360	280	200	185	175	160	150	125	100[4]						
50	885	425	350	275	250	225	215	200	175	150	100[4]					
55	990	495	425	350	335	325	300	275	240	200	175	150				
60	1100	570	510	450	425	400	375	350	315	275	240	200	150			
65	1200	645	585	525	515	500	475	450	415	375	315	275	215	150		
70	1250	695	675	625	615	600	575	550	515	475	425	375	315	250	150	
75	1350	730	745	710	705	695	685	675	640	600	550	500	425	350	275	200

NOTES:	
1	Typical conditions are locations where the road user must use extra time to adjust speed and change lanes in heavy traffic because of a complex driving situation. Typical signs are Merge and Right Lane Ends. For other examples of signs for this condition, see "Condition A, Complex Maneuvers" in Table 4.
2	Typical condition in the warning of a potential stop situation. Typical signs are Stop Ahead, Yield Ahead, Signal Ahead, and Intersection Warning signs. For other examples of signs for this condition, see "Condition B, Stop Condition" in Table 4.
3	Typical conditions are locations where the road user must decrease speed to maneuver through the warned condition. Typical signs are Turn and Curve signs with advisory speed plaques. For other examples of signs for this condtion, see "Condition B, Decelerate to Indicated Advisory Speed" in Table 4.
4	The minimum advance placement distance is listed as 100 feet to provide adequate spacing between signs.

Warning signs should not be placed at distances more than twice the minimum shown in Table 3. If signs must be placed at more than twice the minimum distance they should be accompanied by a supplementary plaque.

Some signs that are listed under the Stop Condition section in Table 4 are marked with a double asterisk **. These signs may be used for a segment of roadway rather than a specific point. If signing for a specific point the sign should be placed at least the minimum distance back from that point. If signing for a segment of roadway, such as a Bear Crossing symbol with a Next 5 Miles plaque, the signs should be placed at each end of the segment, not in advance.

Warning signs that are listed for a stop condition should be placed so that they are visible before a curve, or a steep slope or hill. If a sign that may require a driver to stop suddenly is placed where it is only visible in the middle of a curve or on a steep slope or hill the driver may be adversely affected. Refer to the MMUTCD for the specific use of each sign listed in Table 4.

Advisory speed plaques shall not be used on signs that are listed as a stop condition in Table 4.

In order to inform drivers of changes in the horizontal alignment of a roadway, it may be necessary to place Turn, Curve, Reverse Turn, Reverse Curve, or Winding Road signs in advance of such changes. Some changes in horizontal alignment are obvious and no signs may be needed. Other changes in alignment, however, may be more pronounced or hidden from view. In these cases, advance warning signs may be necessary.

Warning signs that are likely to be used with advisory speed plaques are Turn, Curve, Reverse Turn, Reverse Curve, Winding Road, Hairpin Curve, Truck Rollover, and 270 Degree Loop. Refer to the MMUTCD for the specific use of each sign listed. The sign should be placed as shown in Appendix B-31.[30]

If the Combination Horizontal Alignment Advisory Speed sign (W1-1a and W1-2a) is to be used, it shall be used as a supplement to the advanced Turn or Curve sign and is to be placed at the beginning of the turn or the curve. The advisory speed displayed should be determined using the following recommended engineering practices.

The Advisory Speed (W13-1P) plaque may be used to supplement many warning signs to indicate the advisory speed for a condition.

The use of the Advisory Speed plaque for horizontal curves shall be in accordance with the information shown in Table 5 below. The Advisory Speed plaque shall also be used where an engineering study indicates a need to advise road users of the advisory speed for other roadway conditions. If used, the Advisory Speed plaque shall carry the message XX MPH. The speed displayed shall be a multiple of 5 mph. Except in emergencies or when the condition is temporary, an Advisory Speed plaque shall not be installed until the advisory speed has been determined by an engineering study.

The Advisory Speed plaque shall only be used to supplement a warning sign and shall not be installed as a separate sign installation.

The advisory speed shall be determined by an engineering study that follows established engineering practices.

Among the established engineering practices that are appropriate for the determination of the recommended advisory speed for a horizontal curve are the following:

1. An accelerometer that provides a direct determination of side friction factors.
2. A design speed equation.
3. A traditional ball-bank indicator using the following criteria:
    a. 16 degrees of ball-bank for speeds of 20 mph or less,
    b. 14 degrees of ball-bank for speeds of 25 to 30 mph,
    c. 12 degrees of ball-bank for speeds of 35 mph and higher.

    The 16, 14, and 12 degrees of ball-bank criteria are comparable to the current AASHTO horizontal curve design guidance. Research has shown that drivers often exceed existing posted advisory curve speeds by 7 to 10 mph.
4. Field method for the determination of advisory speed on curves.[31]

The advisory speed should be determined based on free-flowing traffic conditions. Because changes in conditions, such as roadway geometrics, surface characteristics, or

[30] Appendix B-29: Minimum Advance Placement Distance for Curve Signing Required in Condition B: Decelerate to Indicated Advisory Speed
[34] Appendix B-12: Determination of Advisory Speeds on Curves

sight distance, might affect the advisory speed, each location should be evaluated periodically or when conditions change.

For freeway ramps with a speed reduction of 40 MPH or more, a ground mounted warning sign, as required in Table 5, should be supplemented with an overhead warning sign and an exit gore warning sign plaque.[32] A high crash history may also warrant the additional warning signs. Rest area ramps should include a warning sign for speed also.

All horizontal alignment signs shall have reflective panels added to the sign supports. This refers to all signs in the W1 series. Follow the recommendations in 12SP810M.[33] Information on construction can be found in Section 810 of the Standard Specifications for Construction.[34]

Table 5. Horizontal Alignment Sign Selection

TYPE OF HORIZONTAL ALIGNMENT SIGN	DIFFERENCE BETWEEN SPEED LIMIT AND ADVISORY SPEED				
	5 mph	10 mph	15 mph	20 mph	25 mph or more
Turn, Curve, Reverse Turn, Reverse Curve, Winding Road, and Combination Horizontal Alignment/Intersection	Recommended	Required	Required	Required	Required
Advisory Speed Plaque	Recommended	Required	Required	Required	Required
Chevrons and/or Directional large Arrow	Optional	Recommended	Required	Required	Required
Ground Mounted Exit Speed and Ramp Speed on exit ramp	Optional	Optional	Recommended	Required	Required

Note: Required means that the sign and/or plaque shall be used, recommended means that the sign and/or plaque should be used, and optional means that the sign and/or plaque may be used.

CHEVRON ALIGNMENT SIGNS

Chevron Alignment signs are intended as a supplement to the standard delineator treatment and as an alternate or supplement to the One Direction Large Arrow sign. When used, Chevron Alignment signs shall be installed on the outside of curves or turns where run-off-roadway crashes are frequent and additional emphasis on delineation is desired. When a One Direction Large Arrow sign is used in combination with Chevron Alignment signs, the One Direction Large Arrow sign should be placed in line with and

[32] Appendix B-10: Advisory Speed Signs for Freeway Ramps
[33] Michigan DOT Special Provisions
[34] Michigan DOT Standard Specifications for Construction

at approximately a right angle to approaching traffic. The Chevron Alignment signs are to be properly spaced preceding and following the One Direction Large Arrow sign. On non-freeways a One Direction Large Arrow sign should always be used (and placed in target position) when using a series of chevrons.

Chevron alignment signs need to be positioned to provide 500 feet of visibility. At minimum, the motorist must have a continuous view of two signs with low beam headlights along the curved alignment until the change in alignment eliminates the need for Chevron alignment signs. The near edge of the sign should be located a minimum of three feet from the shoulder hinge point, three feet from the face of straight-faced curb, or three feet from the back of guardrail posts or other positive barrier. The three-foot lateral offset is only a minimum dimension. Signs should be offset as far as practical, up to 17 feet from the edge of pavement except when located behind guardrail or other positive barrier where the three-foot offset is desirable.[35]

REFLECTIVE STRIPS ON SIGN POSTS

All W1 series signs, and midblock Pedestrian, Bike, Trail, and School Crossings shall have reflective strips installed on the posts. Stop Ahead and Yield Ahead, if used, and if reflective strips are used on the STOP and YIELD signs[36], shall have reflective strips. The sheeting of the reflective panel must match the background color of the sheeting of the sign mounted on the post.

ADVANCED SNOWMOBILE AND ALL TERRAIN VEHICLE (ATV) CROSSING SIGNS

When requested by a governmental agency, an advance crossing sign should be placed at Michigan Department of Natural Resources and Environment (MDNRE) sanctioned trail crossings with sight distances less than intersection sight distance for the posted speed of the roadway. See the department's Sight Distance Guidelines for further information on intersection sight distance.

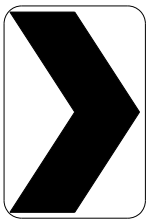
For crossings exhibiting a crash pattern, the Region Traffic and Safety Representative should contact the MDNRE to determine if the crossing can be relocated or modified to address the crash pattern. For those locations exhibiting a crash pattern where no revisions can be made, the advance crossing sign should be placed to inform motorists of possible crossings or entries into the roadway.

Advance warning signs may also be placed on engineering judgment upon a joint review by MDOT and MDNRE. Advance warning signs used may be the Snowmobile warning sign, the ATV warning sign, or if a combined snowmobile and ATV trail, the TRAIL CROSSING text sign.

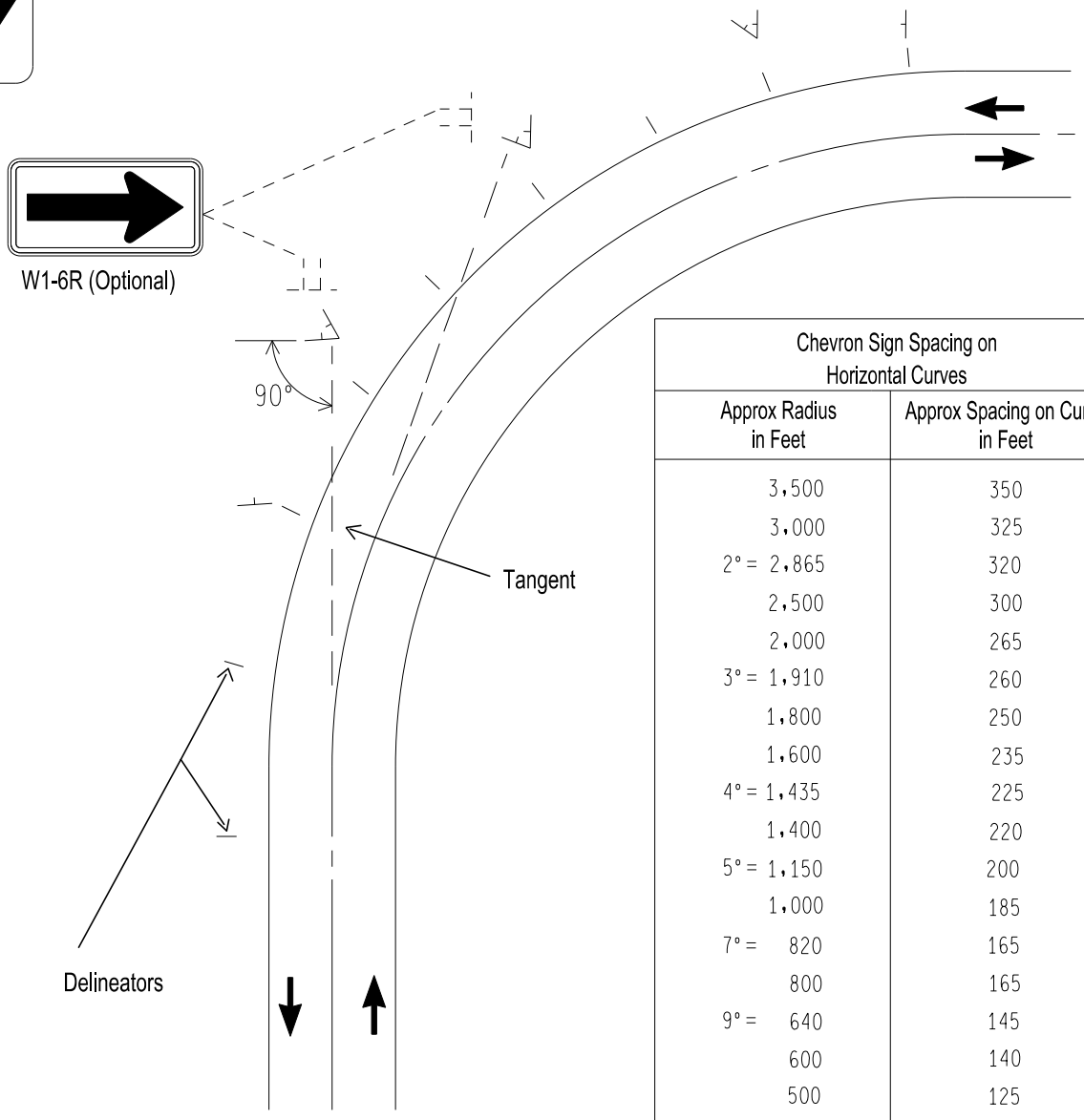
Snowmobile[37] and ATV[38] trail maps are available from the Michigan Department of Natural Resources and Environment website.

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[35] Appendix B-9: Roadway Delineation
[36] SIGN-145: All Way Stop and Cross Traffic Does Not Stop Conditions
[37] DNR Snowmobile Trails
[38] DNR ATV Trails



Chevron alignment sign (W1-8) standard sizes
24" X 30" - Conventional roadways & freeway ramps 20-40 MPH
36" X 48" - Freeway ramps 45-55 MPH & freeway mainline



Chevron Sign Spacing on Horizontal Curves	
Approx Radius in Feet	Approx Spacing on Curve in Feet
3,500	350
3,000	325
2° = 2,865	320
2,500	300
2,000	265
3° = 1,910	260
1,800	250
1,600	235
4° = 1,435	225
1,400	220
5° = 1,150	200
1,000	185
7° = 820	165
800	165
9° = 640	145
600	140
500	125
12° = 480	125
15° = 385	110
300	95
275	90
230	80
30° = 195	70
40° = 145	60
100	40

Notes:

1. The appropriate spacing "S" on the curve is found from the formula $S = 6\sqrt{R-50}$ where "R" is the radius curve in feet.
2. The signs should be spaced evenly.
3. The target arrow replaces a chevron, it is not additional to the chevrons.
4. If a target arrow is used, a target arrow in each direction should be placed first in target position and then the chevrons evenly spaced after that.
5. Do not attach chevron signs to existing sign installation or spaced delineators.

MDOT Michigan Department of Transportation

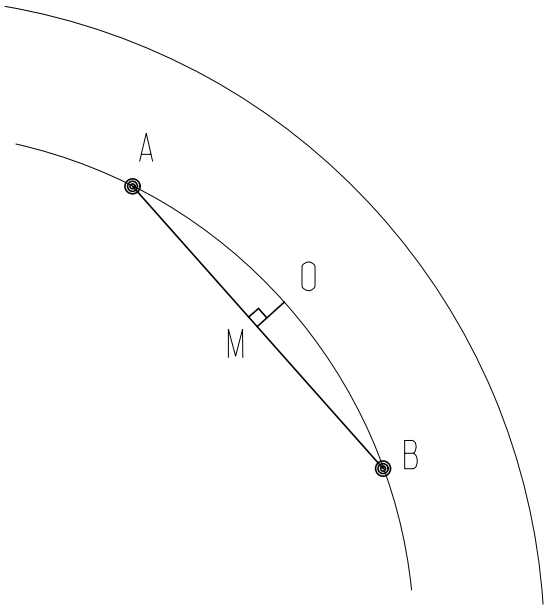
TRAFFIC AND SAFETY

ROADWAY DELINEATION

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Field procedure to determine the radius of curvature (R):

1. Measure 100 ft cord A-B (inside edge of curve)
2. Measure M-O (from midpoint of AB)
3. Read R from table



M-O in ft	R in ft
0.22	5730
0.44	2865
0.66	1910
0.87	1432
1.09	1146
1.31	955
1.53	819
1.75	716
1.97	637
2.18	573
2.40	521
2.62	477
2.82	441
3.00	409
3.22	382
3.45	358
3.67	337
3.93	318
4.15	302
4.36	286
4.58	273
4.80	260
5.02	249
5.24	239

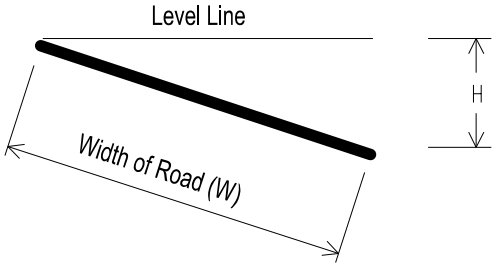
Field procedure to determine rate of superelevation (e):

1. Measure W and H at several points on curve (not in transition areas)

2. Calculate e: $e = \frac{H}{W}$

$$f + (0.01)\, e = \frac{V^2}{15\, R}$$

Where:
e = Superelevation (%)
f = Side friction
V = Velocity (mph)
R = Radius (ft)

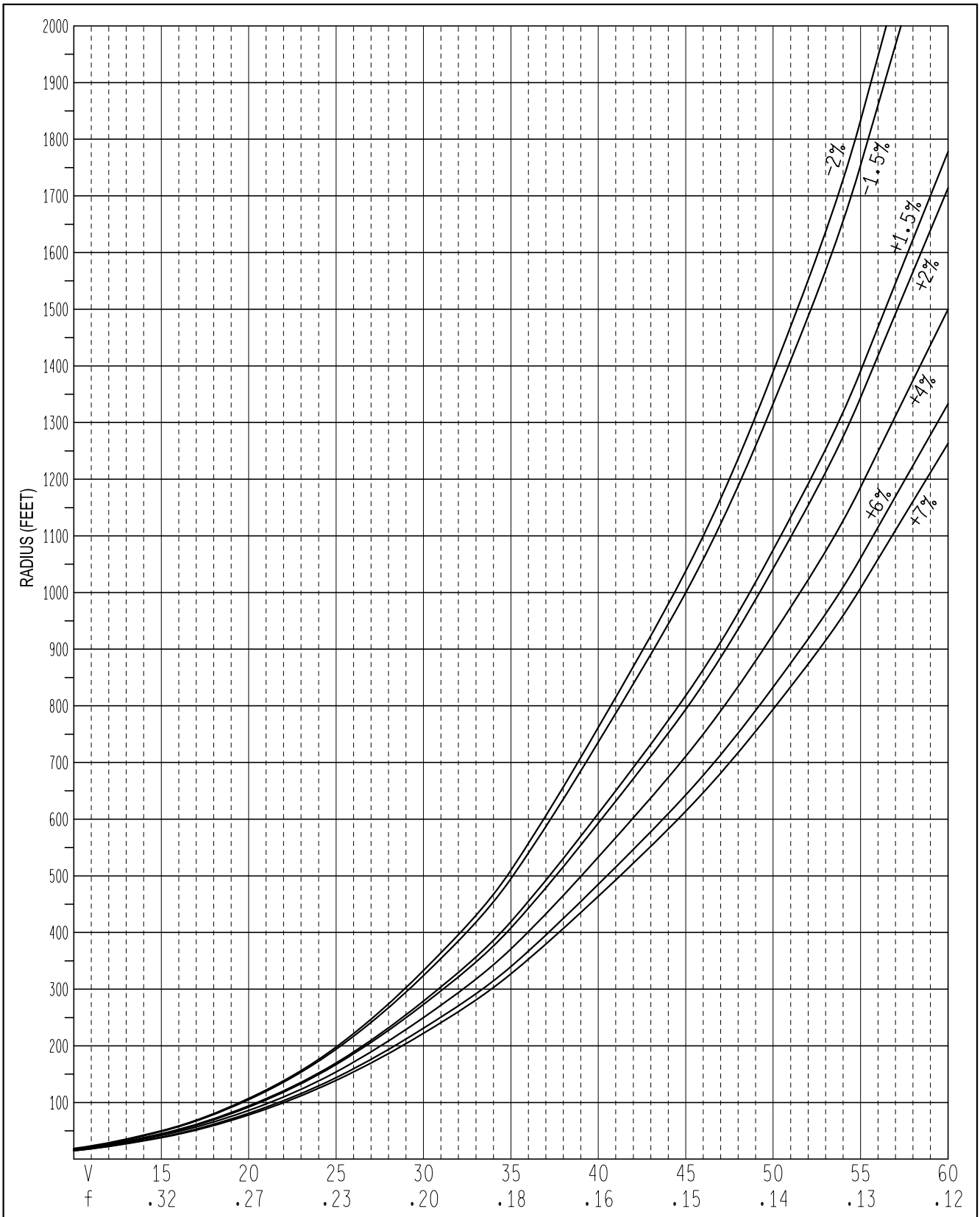


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TRAFFIC AND SAFETY

DETERMINATION OF ADVISORY SPEED ON CURVES

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V = Speed in mph

f = Assumed side friction factor from page 3-25 figure 3-6 of A Policy on Geometric Design of Highways and Streets by AASHTO (2011 Edition).

MDOT Michigan Department of Transportation

TRAFFIC AND SAFETY

DETERMINATION OF ADVISORY SPEED ON CURVES

DRAWN BY: QHN	07/19/2022		B-12
CHECKED BY: AJU	PLAN DATE:		
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